

sorted, and whites and delicates which were least soiled had to be washed first, quickly, followed by dirtier and dirtier items of clothing. Hot water didn't run in taps at the time. Early washing machines (late 1700s) were hand-cranked oddities that none of us would even recognise as a washing machine. They were simple though, and some exist even today in museums. Complexity grew to the point where electricity got involved. First to wash the clothes, then to also heat the water.

In the 1930s, in the US, close to half a million washing machines were being sold a year! The two roller "wringers" were replaced by drum spinning dryers. Today almost every middle-class household has one, varying in complexity from simple cheap and old models that still need manual

innovate and offer new features that others do not have, which consumers deem worthy of trading up for. This approach involves a lot of R&D expenditure, and this is why cutting edge products are more expensive.

Another way this is achieved is to ensure that products don't last forever. There is a sweet-spot that the market tends to find. If brand X's machine dies out too quickly, it earns a bad reputation and has to shut shop. The sweet spot that brands aim for is to have machines last just long enough to not be considered a failure.

Hypothetically, (we're not washing machine designers or market planners to know any better), a rich consumer buying a top-of-the-line machine might be willing to upgrade it every, say, 5 to 7 years, in order to get something with the latest technology. A middle-class

household might want one that lasts, say, 8 to 10 years. This includes acceptable levels of repair in that lifetime of course. For poor households, they'd want a washing machine that lasted forever, they will never get it though. The machines they can afford might be designed to die in just 5 years. Why? Because you want your middle-end and high-end products to be something people aspire to own.

A brand might design a machine for rich people that's capable of performing well even 10 to 12 years after manufacture (despite knowing that it will be only used for 5 to 7 years). They'd want it to spend a little time as a second-hand product used by middle class or poor people, to give them a taste of "the good life". The same brand will design middle-level machines to last less than their flagship models, and the budget models might be designed to die the quickest. Often the price constraints within which the brands work, might itself be the reason for cheaper parts being used, which are prone to failure more quickly than high-end products.

Planned obsolescence can also be achieved by phasing out spares for budget devices older than, say, 5 years, or making spares hard to come by, so

that buying a new machine just seems like a better option to consumers.

If a spare part for your eight-year-old washing machine costs half as much as a new machine does, it's not a compelling option, given that newer machines will promise better features. Or, even if the part is affordable, and can be imported from, say, Taiwan, in just four to six weeks, you start thinking about a new machine that can be delivered to you in under three days... Given these options most people will think, "Oh well, six years was a good run, might as well upgrade now!" or "Six weeks of having to handwash clothes? Not worth the trouble!"

COMPLEXITY

While our example of washing machines might not be relatable to everyone, the examples of phones will definitely resonate with you. Phones are constantly getting more and more complex and compact over the years, and we've come such a long way from our editor's Ericsson GH-337 brick of a device to the latest flagship phones from your favourite manufacturer. Just the removable battery is something that died out long ago. Protecting the phone's inside from moisture and other weather phenomena, came at the cost of being able to swap out a phone's battery when it holds less and less charge over time. No matter what phone you have or what battery, it will eventually degrade and you will find yourself charging more often every day. Of course you cannot open up your own phone to change the battery because the methods of opening every device (without damaging internal components) is different. Add to that the weatherproof sealing that's done and you know you need professional help just to be able to change a battery.

The same applies for any part that needs replacing inside modern phones. You literally need an engineering degree and often schematics of the device. You would also need to have the parts available to you in order to be able to swap them out. Broken screens or broken back glasses of phones are a common occurrence thanks to human error of dropping phones. This means



Ed really wants you to know how hardy his Ericsson GH 337 was

intervention to use wringers, to fully automatic, AI-driven, fuzzy logic wielding robots that are eco friendly in terms of water and electricity usage, but still clean your clothes far better than you could by hand.

Now, with a market that's saturated, with everyone owning a washing machine, there are only two real markets left for brands. The replacements / upgrades market where families with washing machines buy newer models for more features, or replace one that's died after years of service. The second is a very price sensitive market for those emerging from poverty and looking to buy their first machine on a budget.

Brands fight for these two markets, and they do so by attempting to